METRIC-UNIFORM SPECTRAL INEQUALITY FOR THE LAPLACIAN ON MANIFOLDS WITH BOUNDED SECTIONAL CURVATURE

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Abstract. Given a Riemannian manifold M endowed with a smooth metric g satisfying sectional upper and lower curvature bounds, we show an equivalence property between the L^2 norm on M and the L^2 norm on subsets ω satisfying a thickness condition, for functions in the range of a spectral projector. The thickness condition is known to be optimal in this setting. The constant appearing in the equivalence of norms property depends only on the dimension of the manifold, curvature bounds, and frequency threshold of the spectral cutoff but, crucially, not on the injectivity radius.

1. Introduction

1.1. **Metric-uniform spectral inequality.** Let (M, g) be smooth, complete Riemannian manifold. We investigate quantitative unique continuation from a measurable set $\omega \subset M$ for functions with compact (Laplace)-spectral support, a type of result known as a *spectral inequality* in the control theory literature. Starting with the seminal work of Lebeau and Robbiano [LR95], spectral inequalities are related to null-controllability (or equivalently, final-time observability) of the heat equation.

When M has Ricci curvature bounded below, Deleporte and Rouveyrol proved in [DR24] that a necessary condition for ω to support a spectral inequality is that it satisfies some quantitative equidistribution property in the whole manifold, known as *thickness*. In the compact [BM23], flat [EV18, WWZZ19] and hyperbolic [Rou25, DR24] settings, thickness of ω has in turn been shown to imply that it supports a spectral inequality. In this context, the aim of the present article is to prove that the spectral inequality holds from thick subsets in any manifold with bounded sectional curvature, with bounds that depend only on the dimension, the norm of the sectional curvature and the thickness of ω but, crucially, not on the injectivity radius of M (which may tend to zero at infinity). This result implies null-controllability of the heat equation with constants depending only on those geometric quantities.

In order to state our results precisely, let us define a few objects. On (M, g), we denote by $-\Delta_g \geqslant 0$ the Laplacian, also known as the Laplace–Beltrami operator, defined as the Friedrichs extension of the Dirichlet form, see [Lew22]. Since the Laplacian is non-negative and self-adjoint it has a positive square root, to which we associate a spectral measure μ supported on $[0, \infty)$ and forming a resolution of the identity in the sense that for all $u \in \mathrm{L}^2(M)$, $u = \int_0^\infty \mathrm{d}\mu(\lambda)u$. For $\Lambda > 0$, let Π_Λ be the projector onto the spectral window $[0, \Lambda]$ defined via the functional calculus as

$$\Pi_\Lambda u = \int_0^\Lambda \mathrm{d}\mu(\lambda)u.$$

Definition 1.1. *Let (M, g) be a Riemannian manifold and $\omega \subset M$ be measurable. We say that* (M, g, ω) satisfies the spectral inequality *if there exists $C_{\mathrm{spec}} > 0$ such that for all $\Lambda > 0$*

and $u \in \mathrm{L}^2(M)$,

$$\|\Pi_\Lambda u\|_{\mathrm{L}^2(M)} \leq e^{C_{\mathrm{spec}}(\Lambda+1)} \|u\|_{\mathrm{L}^2(\omega)}. \tag{Spec}$$

Thickness of a measurable set ω is, on its face, a purely measure theoretical property.

Definition 1.2. *Let* (M, g) *be a Riemannian manifold,* Vol *its canonical Riemannian measure,* $\omega \subset M$ *measurable, and* $R > 0$, $\varepsilon \in (0, 1)$. *We say that* ω *is* thick at scale R with ratio ε *(or* (R, ε)*-thick) if for all* $x \in M$, *denoting* $B_x(R) \subset M$ *the geodesic ball of center* x *and radius* R,

$$\frac{\mathrm{Vol}(\omega \cap B_x(R))}{\mathrm{Vol}(B_x(R))} \geq \varepsilon. \tag{Thick}$$

We say that ω *is* thick *if there exist some* $R > 0$ *and* $\varepsilon \in (0, 1)$ *such that* ω *is thick at scale* R *with ratio* ε. *In the rest of the article,* (R, ε) *will be called the* thickness parameters.

We aim at addressing the following questions.

1. What is the largest class of manifolds on which the equivalence between (Spec) and (Thick) holds?
2. Can the constant C_{spec} be proved to depend only on geometric parameters of the manifolds (M, g) (along with the thickness parameters (R, ε)) rather than on the metric g itself?

With respect to Question 1, we obtain a (partial) converse to the necessity statement of [DR24], while for Question 2 we prove that the dimension and a bound on the sectional curvature are the only geometric quantities on which the spectral inequality depends. For $d \in \mathbb{N}$ and $\kappa \geqslant 0$, denote by $\mathcal{M}(d, \kappa)$ the set of d-dimensional manifolds whose sectional curvature has norm smaller than κ.

Theorem 1.3. *Let* $d \geq 2$, $\kappa \geqslant 0$ *and* $R, \varepsilon > 0$. *There exists a constant* C_{spec} *such that for any* (R, ε)*-thick subset* ω *of a Riemannian manifold* $(M, g) \in \mathcal{M}(d, \kappa)$ *supports a spectral inequality with constant* C_{spec}.

1.2. **Applications to observability and control theory.** One application of spectral inequalities of the form (Spec) is the controllability and observability of the heat equation.

Definition 1.4. *The* controllability cost *at time* $T > 0$ *for the heat equation is the constant*

$$C_{\mathrm{cont}}(T, M, \omega) = \sup_{u_0, f} \frac{\|f\|_{\mathrm{L}^2((0,T)\times\omega)}}{\|u_0\|_{\mathrm{L}^2(M)}},$$

where the supremum is taken over all initial conditions $u_0 \in \mathrm{L}^2(M) \setminus \{0\}$ *and controls* $f \in \mathrm{L}^2((0, T) \times M)$ *such that the solution of*

$$\begin{cases} \partial_t u - \Delta_g u &= \mathbb{1}_\omega f \quad \text{in } \mathbb{R}_t^+ \times M, \\ u_{|t=0} &= u_0 \quad \text{in } M \end{cases}$$

satisfies $u_{|t=T} = 0$.

The obervability cost *for the heat semigroup at time* $T > 0$ *is the smallest constant* $C_{\mathrm{obs}}(T, M, \omega)$ *such that the final-time observability inequality*

$$\forall u \in \mathrm{L}^2(M), \|e^{T\Delta_g} u\|_{\mathrm{L}^2(M)} \leq C_{\mathrm{obs}} \int_0^T \int_\omega |e^{t\Delta_g} u(x)|^2 d\,\mathrm{Vol}\, dt \tag{Obs}$$

is satisfied.

By duality, controllability and observability are equivalent; moreover, using general functional calculus arguments, both are consequences of the spectral inequality (Spec).

Proposition 1.5. *(* [Cor07, Theorem 2.44], [Mil10, Section 3.2], [WWZZ19]*) Let (M,g) be a complete Riemannian manifold, and $\omega \subset M$ a measurable subset of M such that* (Spec) *holds from ω. Then, ω satisfies* (Obs) *and there exists $c_1 > 0$ such that for every $T > 0$,*

$$C_{\mathrm{cont}}(T) = C_{\mathrm{obs}}(T) \leq \exp(c_1(1 + C_{\mathrm{spec}})^2(1 + \tfrac{1}{T})).$$

In the context of Theorem 1.3, given $d, R, \varepsilon, \kappa$, Proposition 1.5 yields a uniform observability cost for (R,ε)-thick sets on d-dimensional manifolds with curvature bounded by κ.

Deleporte and Rouveyrol proved the following converse statement, which illustrates the sharpness of Theorem 1.3.

Proposition 1.6 ([DR24], Theorem 2)**.** *Let $d \geq 2$, (M,g) be a d-dimensional manifold with Ricci curvature bounded from below and $\omega \subset M$. If $C_{\mathrm{obs}}(T, M, \omega) < +\infty$, then ω is thick.*

Beyond (Obs), we also prove an observability result for the heat equation *at fixed time*.

Theorem 1.7. *Let Let $d \geq 2$, $\kappa \geqslant 0$, and $R, \varepsilon, s > 0$. There exist $C > 0$ and $\alpha \in (0,1)$ such that for any Riemannian manifold $(M,g) \in \mathcal{M}(d,\kappa)$, and any (R,ε)-thick set $\omega \subset M$, for every $u \in \mathrm{L}^2(M,\mathbb{R})$,*

$$\|e^{s\Delta_g} u\|_{\mathrm{L}^2(M)} \leq C \|e^{s\Delta_g} u\|^{\alpha}_{\mathrm{L}^2(\omega)} \|u\|^{1-\alpha}_{\mathrm{L}^2(M)}.$$

In particular, there exists C_1 such that, for every $\delta > 0$,

$$\|e^{s\Delta_g} u\|_{\mathrm{L}^2(M)} \leq C_1 \delta^{-\frac{1-\alpha}{\alpha}} \|e^{s\Delta_g} u\|_{\mathrm{L}^2(\omega)} + \delta \|u\|_{\mathrm{L}^2(M)}.$$

1.3. **Further remarks and comparison to existing results.** The spectral inequality (Spec) has previously been investigated in the non-compact manifold setting in [Mil05] and [RT24]. Our geometric assumption is slightly stronger than the one used in both of these works (bounded sectional curvature rather than a lower bound on Ricci curvature), though we do not need the interpolation inequality assumption of [Mil05] and obtain a high-frequency result, unlike in [RT24]. The sufficiency of thickness for spectral observability is new in such general geometries, and so is the uniform character of C_{spec}. Besides, our proof is still valid in the compact setting, generalizing previous spectral estimate results [BM23, LL21] by specifying the qualitative geometric dependence of the constant.

Theorem 1.7 is inspired by similar pointwise observability results for weakly diffusive [AM22] and Baouendi–Grushin operators [AS25], which stem from similar propagation of smallness arguments.

1.4. **Summary of the proof.** We now sketch the proof of Theorems 1.3 and 1.7. The general scheme borrows from the works of Jerison, Lebeau and Robbiano [LR95, JL99] and their recent adaptations to the non-compact manifold setting [BM25, Rou25, DR24].

Given $(M,g) \in \mathcal{M}(d,\kappa)$, $\Lambda > 0$ and a function $u \in \mathrm{Ran}(\Pi_\Lambda)$, we define, via the functional calculus, a harmonic extension of u by

$$w(t,x) = \frac{\sinh(t\sqrt{-\Delta_g})}{\sqrt{-\Delta_g}} u,$$

which satisfies

$$(\partial_t^2 + \Delta_g) w = 0 \text{ in } \mathbb{R} \times M,$$

as well as boundary conditions suitable to applying elliptic propagation of smallness estimates to w. A similar harmonic extension of $e^{s\Delta_g}u$ is used in the proof of Theorem 1.7.

We then apply Logunov and Malinnikova's propagation of smallness inequalities from codimension 1 subsets for gradients of harmonic functions [LM18, Theorem 5.1] in carefully constructed charts. However, to obtain a uniform local inequality, these charts need to satisfy the following conditions:

(i) the domain of the charts must be the same,
(ii) the metric must have uniform ellipticity and Lipschitz constants in those charts,
(iii) the measure of the sets from which smallness is propagated must be uniformly bounded from below.

We solve issues (i) and (ii) using harmonic coordinates. For the metric to be sufficiently regular in those coordinates we would require at least a bound on the Ricci curvature. However, the construction of harmonic coordinates also requires a lower bound on the local injectivity radius. Besides, point (iii) cannot be straightforwardly implied by (Thick) due to the possibly small denominator $\mathrm{Vol}(B(x,R))$ (for example, $\mathrm{Vol}(B(x,R))$ vanishes asymptotically when x tends towards the end of a cusp in a hyperbolic surface). We solve both issues at once by constructing good charts in a local covering space of the manifold by the exponential map, the *normal cover*, which requires us to strengthen our assumption to bounded sectional curvature.

The norm bound on the sectional curvature enters play in two different ways. First, as a lower bound on sectional curvature which provides a key volume doubling inequality (Proposition 2.4) that is repeatedly used throughout the article, notably for counting arguments. Then, as an upper bound on the sectional curvature which plays a role in the construction of the aforementioned harmonic coordinates.

We note that replacing the Logunov–Malinnikova inequality with [LBM25, Theorem 2.2] generalises our result to operators $H = -\Delta_g + V$ with potentials $V \in \mathrm{L}^\infty(M)$ with the caveat that there is an additional dependence of C_{spec} on $\|V\|_{\mathrm{L}^\infty}$.

The article is divided as follows: in Section 2, we recall geometric preliminaries, and define and investigate the $\mathcal{M}(d,\kappa)$ class of manifolds and the uniform properties of their normal covers. In Section 3, we prove the local elliptic estimates based on the Logunov–Malinnikova inequality and extend them uniformly within the class $\mathcal{M}(d,\kappa)$. Section 4 contains the conclusion of the proof of Theorems 1.3 and 1.7.

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2. Geometric preliminaries

2.1. **Geometric quantities and notation.** Let M be a d-dimensional smooth manifold. A Riemannian metric g on M is an assignment, for every $x \in M$, of a positive definite bilinear form on the tangent space T_xM, that is a positive section of $T^*M \otimes T^*M$. Locally, the metric

g has the coordinate expression

$$g(x) = \sum_{j,k} g_{jk}(x) \mathrm{d}x^j \otimes \mathrm{d}x^k,$$

where $\{\mathrm{d}x^j : 1 \leqslant j \leqslant d\}$ is a smoothly varying basis for the cotangent space. We denote by $g^{jk}(x)$ the local coordinate expression for the inverse of the metric and regularly omit the dependence of the metric (and of other tensors) on the base point x. While the metric is an intrinsic object, in the sense that the inner product on TM is independent of the choice of local coordinates, its regularity can only be inspected in coordinates and is defined in a chart as the regularity of the coefficients g_{jk}. The main control we need is given as follows, of course defined for the expression of a metric in a chart.

Definition 2.1. *Let g be a Riemannian metric defined on some open set $\Omega \subset \mathbb{R}^d$. The* biLipschitz constant $\mathrm{Lip}(g)$ *is defined as*

$$\mathrm{Lip}(g) := \max\left\{\|g\|_{\mathrm{W}^{1,\infty}(\Omega)}, \|g^{-1}\|_{\mathrm{L}^\infty}(\Omega)\right\}.$$

If g is defined on a manifold, its biLipschitz constant is the infimum over all coverings of M by charts of the supremum of the biLipschitz constants read in each of those charts.

For $x \in M$ and $r > 0$, we denote the ball of radius r around x by $B_x(r)$. Since we will repeatedly use alternate between balls in M and balls in T_xM we denote the ball of radius r around $0 \in T_xM$ by $\widetilde{B}_x(r)$. In particular, this data gives us one example of a geometrically defined chart around a point.

Definition 2.2. *Given a Riemannian manifold with Lipschitz metric (M, g) and $x \in M$, the* exponential map $\exp_x : T_xM \to M$ *assigns to $v \in T_xM$ the position $\exp_x(v)$ at time 1 of the geodesic with initial data (x, v) at time 0. The* injectivity radius *at x is the maximal $r > 0$ such that the restriction $\exp_x : \widetilde{B}_x(r) \to B_x(r)$ is injective.*

The Laplacian on M, also known as the Laplace–Beltrami operator, is given in local coordinates as

$$\Delta_g u = \frac{1}{\sqrt{|\det g|}} \sum_{j,k} \partial_j \left(\sqrt{|\det g|} g^{jk} \partial_k\right),$$

in other words it is represented locally as an elliptic operator in divergence form, and any theorem stated in Euclidean space for such an operator can be applied wholesale in charts to the Laplace–Beltrami operator.

Two important intrinsic geometric quantities associated with the metric are the *Ricci curvature* Ric and the *sectional curvature* K. Since we will not be working directly with the expression of those curvatures, but rather with quantities explicitly controlled by bounds on the curvature, we neither provide a precise definition nor describe the difference between both, other than observing that bounds on the sectional curvature imply bounds on the Ricci curvature. We use bounds on the sectional curvature to define classes of manifolds without referring to local charts, we will point out which of our lemmas only require bounds on the Ricci curvature.

Definition 2.3. *Let $\kappa \in [0, +\infty)$ and $d \geq 1$. We denote by*

$$\mathcal{M}(d, \kappa) := \{(M, g) : \dim(M) = d \text{ and } \|K\|_{\mathrm{L}^\infty} \leqslant \kappa\}$$

the set of d-dimensional manifolds with (sectional) curvature bounded by κ. Note that elements of $\mathcal{M}(d, \kappa)$ are neither assumed to be compact nor connected.

2.2. **Volume and density bounds.** Manifolds of bounded curvature satisfy a volume doubling inequality, which in turn implies bounds on the density of uniformly discrete sets. All results in this subsection depend, in fact, only on lower bounds for the Ricci curvatures. The doubling inequality can be read directly off the Bishop–Gromov inequality [Gro07, Lemma 5.3.bis], and we state it for convenience of the reader.

Proposition 2.4. *Let $\kappa > 0$ and $d \in \mathbb{N}$. There is $C > 0$ such that for every $(M, g) \in \mathcal{M}(d, \kappa)$, $x \in M$ and $R > r > 0$, we have the inequality*

$$\mathrm{Vol}(B_x(R)) \leq C \left(\frac{R}{r}\right)^d e^{(d-1)\sqrt{\kappa}(R-r)} \mathrm{Vol}(B_x(r)). \tag{2.1}$$

Proposition 2.4 allows us to control the density of discrete sets in M.

Proposition 2.5. *Let $d \in \mathbb{N}$, $\kappa > 0$. There is $C > 0$ such that for every $r, R > 0$, any $M \in \mathcal{M}(d, \kappa)$, and every r-separated set $\mathbf{Y} \subset M$,*

$$\#(\mathbf{Y} \cap B_x(R)) \leqslant c \left(\frac{R}{r}\right)^d e^{2(d-1)\sqrt{\kappa}R}$$

for all $x \in M$.

Proof. Let $r > 0$ and $\mathbf{Y} \subset M$ be an r-separated set. For all $y, y' \in \mathbf{Y}$,

$$y \neq y' \quad \Longrightarrow \quad B_y(r/2) \cap B_{y'}(r/2) = \varnothing,$$

so that for any $x \in M$ and $R > 0$

$$\sum_{y \in \mathbf{Y} \cap B_x(R)} \mathrm{Vol}(B_y(r/2)) \leqslant \mathrm{Vol}(B_x(R + r/2)) \leqslant C e^{(d-1)\sqrt{\kappa}\frac{r}{2}} \mathrm{Vol}(B_x(R)). \tag{2.2}$$

On the other hand, $B_x(R) \subset B_y(2R)$ for every $y \in \mathbf{Y} \cap B_x(R)$. Thus,

$$\mathrm{Vol}(B_x(R)) \leqslant \min_{y \in \mathbf{Y} \cap B_x(R)} \mathrm{Vol}(B_y(2R)) \leqslant C \left(\frac{R}{r}\right)^d e^{(d-1)\sqrt{\kappa}(2R - \frac{r}{2})} \min_{y \in \mathbf{Y} \cap B_x(R)} \mathrm{Vol}(B_y(r/2)). \tag{2.3}$$

Combining (2.2) and (2.3), we get that, up to maybe choosing a larger C,

$$\#(\mathbf{Y} \cap B_x(R)) \leqslant C \left(\frac{R}{r}\right)^d e^{2(d-1)\sqrt{\kappa}R}.$$

□

We note that on spaces that satisfy the doubling inequality (2.1) the thickness property of Definition 1.2 is well-behaved, in the sense that if a set is thick at scale R_0 it is also thick at any scale $R > 0$. Furthermore, if M has no compact connected component, thickness of a subset $\omega \subset M$ is invariant by removal of any compact set in M.

2.3. **Normal covers and good coordinate systems.** We now aim at constructing local coordinate charts that are uniformly controlled within the classes $\mathcal{M}(d, \kappa)$. In particular, the aim is that

(1) the metric is controlled only by d and κ in those charts,
(2) the size of the neighbourhoods on which the charts map depends only κ.

The injectivity radius is conspicuous by its absence from this list of requirements. Indeed, if we allow dependence on the injectivity radius as well there are many standard choices of coordinate charts which could satisfy our requirements depending on the level of regularity required, for instance harmonic coordinates or the "almost linear coordinates" developed by Jürgen Jost [JK82, Jos84]. These coordinates can only be defined on a neighbourhood whose size is bounded by the injectivity radius. In the class of examples we consider, the injectivity radius is of course everywhere positive, however it may not be bounded away from zero, for instance in the case of a hyperbolic manifold with cusps.

In order to address this difficulty, we will instead use harmonic coordinates on a local cover of M around x; the *normal cover*, and it is on this cover that we will perform our analyses. The injectivity radius of the normal cover depends only on κ, so that the harmonic coordinates will be defined on balls whose radius is uniformly controlled by κ. The price to pay is that the number of pre-images of a point in the normal cover may grow as the injectivity radius goes to zero, but it does so in a quantifiable way which lets us control estimates appropriately.

The exponential map provides a cover of the connected component of M containing x by T_xM, whose metric is given by the pullback $\exp_x^* g$. However, this metric is usually not well-behaved on the whole of T_xM, and it is for this reason that we will define it on a smaller ball. The first step is to show that there is a ball in T_xM in which the injectivity radius of $\exp_x^* g$ is uniformly controlled.

Proposition 2.6. *Let $\kappa > 0$. There is $L > 0$ such that for every $d \in \mathbb{N}$, $M \in \mathcal{M}(d, \kappa)$ and $x \in M$, the restriction*

$$\exp_x : B_{T_xM}(0, \tfrac{\pi}{2\sqrt{\kappa}}) \to M$$

is locally bi-L-Lipschitz. Furthermore, in the metric $\exp_x^* g$, every $v \in B_{T_xM}(0, \frac{\pi}{4\sqrt{\kappa}})$ has injectivity radius larger than $\frac{\pi}{4\sqrt{\kappa}}$.

Proof. BiLipschitzness follows from Rauch's comparison theorem [Rau51], (a modern exposition is found in [Jos84, Lemma 2.2.1]). By [Jos84, Lemma 2.4.2], $B_{T_xM}(0, \frac{\pi}{2\sqrt{\kappa}})$ is geodesically convex under the metric pulled back by the exponential map, meaning that any two points are joined by a unique geodesic segment, directly implying that the injectivity radius is bounded below by $\frac{\pi}{4\sqrt{\kappa}}$ on the ball of radius $\frac{\pi}{4\sqrt{\kappa}}$. □

While the exponential map is itself biLipschitz, it does not provide us with uniform control over the biLipschitz constant of g. As first noticed by DeTurck–Kazdan [DK81], the right choice of coordinates for optimal control of the regularity are *harmonic coordinates*. The following proposition collates statements found in the discussion before Lemma 2.8.1, and in Lemma 2.8.2 of [Jos84] applied to T_xM.

Proposition 2.7. *Let $\kappa > 0$ and $d \in \mathbb{N}$. There is $L, \rho > 0$ and $r_\kappa > 0$, called the* uniform Lipschitz harmonicity radius *so that for every $M \in \mathcal{M}(d, \kappa)$ and $x \in M$, there is a harmonic $\varphi : \widetilde{B}_x(r_\kappa) \to \widetilde{B}_x(\frac{\pi}{4\sqrt{\kappa}})$, diffeomorphic on its image so that $\widetilde{g} := \varphi^* \exp_x^* g$ is biLipschitz with $\mathrm{Lip}(\widetilde{g}) \leqslant L$. Moreover, $\widetilde{g}$ is the canonical metric at 0, and the image of φ contains $B_{T_xM}(0, \rho)$.*

Of course, the harmonic radius usually depends on the injectivity radius of the manifold, but as shown in Proposition 2.6, the injectivity radius on the tangent space can be bounded below in terms of the curvature. The normal cover at a point is simply a ball in the tangent space whose radius is slightly smaller than the harmonic radius.

Definition 2.8. *Let $\kappa > 0$, $d \in \mathbb{N}$, $M \in \mathcal{M}(d,\kappa)$, and $x \in M$. The* normal cover *of M at x is $\widetilde{M}_x := \widetilde{B}_x(r_\kappa)$, endowed with the pullback metric $\widetilde{g} := \varphi^* \exp_x^* g$ expressed in harmonic coordinares, and natural projection $p_x : \widetilde{M}_x \to M$. For every quantity Q defined in a neighbourhood of x (e.g. a subset, a function, the metric, etc.) we denote $\widetilde{Q}$ to be the lift to $\widetilde{M}_x$ of Q,*

While the injectivity radius of $\widetilde{M}_x$ is bounded away from zero independently of x, the degree of the covering $\widetilde{M}_x$ grows unboundedly as x varies in a region where the injectivity radius of M goes to zero. It turns out that as long as the projection of balls centred at 0 in $\widetilde{M}_x$ is distributed uniformly enough on the base space we can recover enough control. In order to make this quantitative, we introduce the following two quantities.

Definition 2.9. *For any $x \in M$ and $r \in (0, \frac{r_\kappa}{4})$, we define the upper pre-image counting function*

$$N(x,r) := \max_{y \in B_x(r)} \#(p_x^{-1}(y) \cap \widetilde{B}_x(r)) \tag{2.4}$$

and the lower pre-image counting function

$$n(x,r) := \min_{y \in B_x(r)} \#(p_x^{-1}(y) \cap \widetilde{B}_x(4r)) \tag{2.5}$$

Proposition 2.10. *For every $\kappa > 0$ and $d \in \mathbb{N}$ there is $C > 0$ so that the pre-image counting functions are equivalent in the sense that.*

$$N(x,r) \leqslant n(x,r) \leqslant C e^{3(d-1)\sqrt{\kappa} r} N(x,r),$$

Furthermore, the pre-image counting functions satisfy the doubling estimates

$$N(x,2r) \leqslant C e^{7(d-1)\sqrt{\kappa} r} N(x,r) \qquad \text{and} \qquad n(x,2r) \leqslant C e^{7(d-1)\sqrt{\kappa} r} N(x,r).$$

Proof. Consider $y, z \in B_x(r)$ such that y is maximizing in (2.4) and z is minimizing in (2.5). Let γ be a path of length at most $2r$ connecting y and z and staying in $B_x(r)$. The lifts $\widetilde{\gamma}$ of γ have the same length and connect injectively preimages $\widetilde{y} \in p_x^{-1}(y)$ and $\widetilde{z} \in p_x^{-1}(z)$. Assuming that $\widetilde{y} \in \widetilde{B}_x(r)$, the triangle inequality gives us $\widetilde{z} \in \widetilde{B}_x(4r)$, so that indeed $N(x,r) \leqslant n(x,r)$. Now, we observe that taking $y = x$ in (2.4) and (2.5) yields

$$n(x,r) \leqslant \frac{\mathrm{Vol}(\widetilde{B}_x(4r))}{\mathrm{Vol}(B_x(r))} \quad \text{and} \quad N(x,r) \geqslant \frac{\mathrm{Vol}(\widetilde{B}_x(r))}{\mathrm{Vol}(B_x(r))}. \tag{2.6}$$

Combining (2.6) and Proposition 2.4, there is $C > 0$ so that

$$n(x,r) \leqslant \frac{\mathrm{Vol}(\widetilde{B}_x(4r))}{\mathrm{Vol}(\widetilde{B}_x(r))} N(x,r) \leqslant C e^{3(d-1)\sqrt{\kappa} r} N(x,r).$$

Similarly, we obtain

$$N(x,2r) \leqslant n(x,2r) \leqslant \frac{\mathrm{Vol}(\widetilde{B}_x(8r))}{\mathrm{Vol}(\widetilde{B}_x(r))} N(x,r) \leqslant C e^{7(d-1)\sqrt{\kappa} r} N(x,r),$$

providing the doubling estimates claimed. □

Remark 2.11. *The sectional curvature bound was only really used in the Rauch comparison theorem to obtain a lower bound on the injectivity radius of the normal cover. As indicated in* [HH97, Theorem 6]*, the uniform Lipschitz harmonic radius can be bounded below in terms of*

the L^∞ *norm of the Ricci curvature and the injectivity radius, so that one can replace sectional curvature with Ricci curvature if there is an a priori lower bound on the injectivity radius of the normal cover. Note, however, that by the same Theorem, we cannot extend this to lower bounds on the Ricci curvature, as that would only guarantee bounds on the* $\mathrm{C}^{0,\alpha}$ *norm of metric in harmonic coordinates, which is not sufficient to later on apply the Logunov–Malinnikova propagation of smallness estimate.*

We conclude this section with a few remarks concerning *a priori* regularity of the metric.

The works [Rau51, JK82, Jos84], and most of the literature built thereupon, assume the Riemannian metric g to be smooth in the original chart.

By density, all arguments laid down in this section extend directly to the case of C^2 metrics with bounded sectional curvature: at least locally, these metrics are C^2 close to smooth metrics with a (slightly worse) bound on the sectional curvature. Then, geodesic rays have a Lipschitz dependence on C^2 variations on the metric, and therefore Proposition 2.6 extends to this situation; in conclusion, our construction of normal coverings applies to this case. Then, one can use again density arguments: after a normal covering, there exists a closeby smooth metric with bounded curvature, and in harmonic coordinates for the smoothed metric are good enough, the initial metric will be controlled in the Lipschitz topology.

More generally, let g_1 be a smooth metric with bounded sectional curvature, and let g_2 be a Lipschitz metric which is *comparable* in the sense that $cg_1 \leq g_2 \leq Cg_1$ everywhere for some $0 < c < C$, and which satisfies $\|\nabla_{g_1} g_2\|_{\mathrm{L}^\infty} < \infty$, a family of charts on coverings for which g_1 is controlled in the Lipschitz topology will also control g_2 in the Lipschitz topology. In conclusion, our main results will also apply to g_2 (notice that the thickness condition is preserved by changing g_1 for g_2).

Unfortunately, there does not seem to be an easy way to characterise Lipschitz metrics satisfying this condition for some g_1, as there is no notion of sectional curvature at this regularity. A necessary condition is the volume doubling estimate, generalising Proposition 2.4.

3. Elliptic tools

This section collects the essential tools towards the proof of Theorem 1.3. First, we recall some classical interior estimates for harmonic functions. Then, we obtain a propagation of smallness result from the famous Logunov–Malinnikova theorem.

Lemma 3.1. *Let* $U \Subset \Omega \Subset \mathbb{R}^d$ *and* $L > 0$. *There is* $C > 0$ *so that for every metric* g *on* Ω *with* $\mathrm{Lip}(g) < L$ *and any* Δ_g*-harmonic functions* $w : \Omega \to \mathbb{R}$ *we have the bound*

$$\|w\|_{\mathrm{C}^1(U)} \leqslant C \|w\|_{\mathrm{L}^2(\Omega)}.$$

Proof. Let U_1 be such that $U \Subset U_1 \Subset \Omega$. It follows from [GT01, Theorem 9.11] that for all $p \in (1, \infty)$,

$$\|w\|_{\mathrm{W}^{2,p}(U_1)} \lesssim_{U,\Omega,p,d,\mathrm{Lip}(g)} \|w\|_{\mathrm{L}^p(\Omega)}, \tag{3.1}$$

and, applying the Sobolev embedding theorem with $p = 2$ to U_1 to see that

$$\|w\|_{\mathrm{L}^{\frac{2d}{d-4}}(U_1)} \lesssim_{U,\Omega,d,\mathrm{Lip}(g)} \|w\|_{\mathrm{L}^2(\Omega)}. \tag{3.2}$$

Let $k = \lceil \frac{d-2}{4} \rceil$, and consider a sequence of sets such that $U \Subset U_k \Subset \ldots \Subset U_1 \Subset \Omega$. Applying recursively inequalities (3.1) and (3.2) to pairs U_j, U_{j-1} we have that for some $q > d$,

$$\|w\|_{\mathrm{L}^q(U_k)} \lesssim_{U,\Omega,d,\mathrm{Lip}(g)} \|w\|_{\mathrm{L}^2(\Omega)}.$$

Applying (3.1) one last time between U and U_k, but this time followed by the Morrey embedding theorem, then reveals

$$\|w\|_{\mathrm{C}^1(U)} \lesssim_{U,\Omega,d,\mathrm{Lip}(g)} \|w\|_{\mathrm{L}^2(\Omega)}.$$

□

We now deduce appropriate propagation of smallness estimates from those of Logunov–Malinnikova [LM18]. We start by deducing a local L^2 estimate.

Proposition 3.2. *Let $d \geq 2$ and let $\Omega \Subset U \Subset \mathbb{R}^d$, and $T, \varepsilon, L > 0$. There exists $C > 0$ and, $\alpha \in (0,1)$ such that every locally Lipschitz metrics g on U with $\mathrm{Lip}(g) < L$, and all pairs $u : \Omega \to \mathbb{R}$ and $w : U \times [-T,T] \to \mathbb{R}$ solving*

$$\begin{cases} (\Delta_g + \partial_t^2) w = 0 & \text{in } U \times [-T,T] \\ w(x,0) \equiv 0 & \text{and } \partial_t w(x,0) = u(x) \end{cases}$$

satisfy

$$\|u\|_{\mathrm{L}^2(\Omega)} \leq C \|u\|^{\alpha}_{\mathrm{L}^2(\omega)} \|w\|^{1-\alpha}_{\mathrm{L}^2(U\times[-T,T])}$$

for every $\omega \subset \Omega$ such that $\mathcal{H}^d(\omega) > \varepsilon$.

Proof. In this proof, we abuse notation by using the same letter for $U \subset \mathbb{R}^d$ and $U \times \{0\} \subset \mathbb{R}^d_x \times \mathbb{R}_t$. On $\mathbb{R}^d_x \times \mathbb{R}_t$ we put the metric $g' = g_x + \mathrm{d}t^2$. We note that since w is constant on U, we have that $|u(x)| = |\nabla w(x,0)|$ for all $x \in U$ and as such it is that quantity that we estimate. In particular, we assume that $\nabla w \not\equiv 0$ on Ω, as otherwise the result holds trivially. Let V be such that $\Omega \Subset V \Subset U$ and let $\omega \subset \Omega$ be such that $\mathcal{H}^d(\omega) > \varepsilon$. For $a > 0$ put $E_a = \{x \in \omega : |\nabla w(x,0)| \geqslant a\}$ so that by Chebyshev's inequality,

$$\mathcal{H}^d(E_a)^{1/2} \leqslant \frac{1}{a} \|\nabla w\|_{\mathrm{L}^2(\omega)}. \tag{3.3}$$

If $\mathcal{H}^d(E_a) < \frac{\varepsilon}{2}$, the Logunov–Malinnikova propagation estimate [LM18, Theorem 5.1] applied to $\omega \setminus E_a \Subset \Omega \Subset V$ (which then has measure larger than $\frac{\varepsilon}{2}$) leads us to

$$\|\nabla w\|_{\mathrm{L}^\infty(\Omega)} \leqslant C \|\nabla w\|^{\alpha}_{\mathrm{L}^\infty(\omega\setminus E_a)} \|\nabla w\|^{1-\alpha}_{\mathrm{L}^\infty(V)} \leqslant C a^{\alpha} \|\nabla w\|^{1-\alpha}_{\mathrm{L}^\infty(V\times[-T,T])},$$

or, rearranging,

$$a^{\alpha} \geqslant \frac{\|\nabla w\|_{\mathrm{L}^\infty(\Omega)}}{C \|\nabla w\|^{1-\alpha}_{\mathrm{L}^\infty(V\times[-T,T])}}. \tag{3.4}$$

Contrapositively, if a doesn't satisfy inequality (3.4) then $\mathcal{H}^d(E_a) > \frac{\varepsilon}{2}$. Inserting this assumption in (3.3) and taking powers of α on both sides gives us

$$\|\nabla w\|_{\mathrm{L}^\infty(\Omega)} \leqslant C \left(\frac{2}{\varepsilon}\right)^{\frac{\alpha}{2}} \|\nabla w\|^{\alpha}_{\mathrm{L}^2(\omega)} \|\nabla w\|^{1-\alpha}_{\mathrm{L}^\infty(V\times[-T,T])}.$$

By definition, then following from Lemma 3.1, we see that

$$\|\nabla w\|_{\mathrm{L}^\infty(V\times[-T,T])} \leqslant \|w\|_{\mathrm{C}^1(V\times[-T,T])} \lesssim \|w\|_{\mathrm{L}^2(U\times[-T,T])},$$

whence the simple application of Hölder's inequality

$$\|u\|_{\mathrm{L}^2(\Omega)} = \|\nabla w\|_{\mathrm{L}^2(\Omega)} \leqslant \mathrm{Vol}(\Omega)^{\frac{1}{2}} \|\nabla w\|_{\mathrm{L}^\infty(\Omega)}$$

completes the proof. □

In the remainder of this section, we extend Proposition 3.2 to manifolds of bounded curvature in $\mathcal{M}(d,\kappa)$. We proceed in three steps, starting with a local estimate in small enough balls then extending to balls of any fixed radius via a covering argument. Finally, we extend the statement to any $M \in \mathcal{M}(d,\kappa)$ under the assumption that ω is (R,ε)-thick inside M. Recall the conventions from Definition 2.8. Namely, given $(M,g) \in \mathcal{M}(d,\kappa)$, the normal cover at $x \in M$ is denoted $\widetilde{M}_x$, we write $B_x(r) = B_M(x,r)$, and for any quantity s associated to M (be it a function, a subset, the metric, etc.), we denote $\widetilde{s}$ the associated quantity on $\widetilde{M}_x$.

Proposition 3.3. *Let $d \geq 2$, $\kappa, \varepsilon, T > 0$, and $r \in (0, \frac{r_\kappa}{4})$. There exist $C > 0$ and $\alpha \in (0,1)$ so that every $(M,g) \in \mathcal{M}(d,\kappa)$, $x \in M$, and pairs $u : B_x(\frac{r_\kappa}{4}) \to \mathbb{R}$ and $w : B_x(\frac{r_\kappa}{4}) \times [-T,T]$ solving*

$$\begin{cases} (\Delta_g + \partial_t^2) w = 0 & \text{in } B_x(\frac{r_\kappa}{4}) \times [-T,T] \\ w(y,0) \equiv 0 & \text{and } \partial_t w(y,0) = u(y) \end{cases}$$

satisfy the estimate

$$\|u\|_{\mathrm{L}^2(B_x(r))} \leq C \|u\|^{\alpha}_{\mathrm{L}^2(\omega)} \|w\|^{1-\alpha}_{\mathrm{L}^2(B_x(\frac{r_\kappa}{4}) \times [-T,T])}$$

for every $\omega \subset B_x(r)$ such that $\mathrm{Vol}(\omega) > \varepsilon \mathrm{Vol}(B_x(r))$.

Proof. Let $\omega \subset B_x(r)$ be such that $\mathrm{Vol}(\omega) > \varepsilon \mathrm{Vol}(B_x(r))$. By Proposition 2.10,

$$\begin{aligned} \mathrm{Vol}(\widetilde{\omega} \cap \widetilde{B}_x(4r)) &\geqslant \mathrm{Vol}(\omega) n(x,r) \\ &\geqslant \varepsilon \mathrm{Vol}(B_x(r)) N(x,r) \\ &\geqslant \varepsilon \mathrm{Vol}(\widetilde{B}_x(r)). \end{aligned}$$

Proposition 2.7 tells us that $\mathrm{Vol}(\widetilde{B}_x(r))$ is bounded away from zero and that $\mathrm{Lip}(\widetilde{g})$ is bounded, depending only on r, κ and d. Applying Proposition 3.2 with $\Omega = \widetilde{B}_x(4r) \Subset U = \widetilde{B}_x(\frac{r_\kappa}{4})$ we obtain $C > 0$ and $\alpha \in (0,1)$ so that

$$\|\widetilde{u}\|_{\mathrm{L}^2(\widetilde{B}_x(4r))} \leq C \|\widetilde{u}\|^{\alpha}_{\mathrm{L}^2(\widetilde{\omega})} \|\widetilde{w}\|^{1-\alpha}_{\mathrm{L}^2(\widetilde{B}_x(\frac{r_\kappa}{4}) \times [-T,T])}.$$

Now, a simple counting argument combined with Hölder's inequality gives on the one hand

$$n(x,r)^{\frac{1}{2}} \|u\|_{\mathrm{L}^2(B_x(r))} \leq \|\widetilde{u}\|_{\mathrm{L}^2(\widetilde{B}_x(4r))}$$

whilst on the other hand

$$\|\widetilde{u}\|_{\mathrm{L}^2(\widetilde{\omega})} \leq N(x, \tfrac{r_\kappa}{4})^{\frac{1}{2}} \|u\|_{\mathrm{L}^2(\omega)} \quad \text{and} \quad \|\widetilde{w}\|_{\mathrm{L}^2(\widetilde{B}_x(\frac{r_\kappa}{4}))} \leq N(x, \tfrac{r_\kappa}{4})^{\frac{1}{2}} \|w\|_{\mathrm{L}^2(B_x(\frac{r_\kappa}{4}) \times [-T,T])},$$

and therefore

$$\|u\|_{\mathrm{L}^2(B_x(r))} \leqslant C \left(\frac{N(x, \frac{r_\kappa}{4})}{n(x,r)} \right)^{\frac{1}{2}} \|u\|^{\alpha}_{\mathrm{L}^2(\omega)} \|w\|^{1-\alpha}_{\mathrm{L}^2(B_x(\frac{r_\kappa}{4}) \times [-T,T])}.$$

Applying repeatedly Proposition 2.10, we find that the ratio $\frac{N(x,\frac{r_\kappa}{4})}{n(x,r)}$ is bounded depending only on r, d, κ, which concludes the proof. □

We now propagate this inequality from small to large balls.

Proposition 3.4. *Let $d \geq 2$ and $\kappa, R, \varepsilon, T > 0$ and put $R' = R + r_\kappa$. There exist $C > 0$ and $\beta \in (0,1)$ so that for every $(M,g) \in \mathcal{M}(d,\kappa)$, $x \in M$ and pairs $u : B_x(R) \to \mathbb{R}$ and $w : B_x(R) \times [-T,T] \to \mathbb{R}$ solving*

$$\begin{cases} (\Delta_g + \partial_t^2) w = 0 & \text{in } B_x(\frac{\pi}{8\sqrt{\kappa}}) \times [-T,T] \\ w(y,0) \equiv 0 & \text{and } \partial_t w(y,0) = u(y) \end{cases}$$

we have the estimate

$$\|u\|_{\mathrm{L}^2(B_x(R))} \leq C \|u\|^{\beta}_{\mathrm{L}^2(\omega)} \|w\|^{1-\beta}_{\mathrm{L}^2(B_x(R') \times [-T,T])}$$

for every $\omega \subset B_x(R)$ such that $\mathrm{Vol}(\omega) > \varepsilon \mathrm{Vol}(B_x(R))$.

Proof. Let $r = \frac{r_\kappa}{16}$ and $\mathbf{Y}$ be a maximal $r/2$-separated set in $B_x(R)$, in particular $B_x(R)$ is covered by balls centred on elements of $\mathbf{Y}$ of radius $r/2$. Let $\mathbf{G}$ be the graph whose set of vertices is $\mathbf{Y}$, and where the edges are given by the relation

$$y \sim y' \iff B_y(r) \cap B_{y'}(r) \text{ contains a ball of radius } r/2.$$

Since $\mathbf{Y}$ is a maximal $r/2$-separated set in the connected set $B_x(R)$, $\mathbf{G}$ is a connected graph, and by Proposition 2.5, the cardinality of its vertex set is bounded above by a constant N depending only on r, κ, R and d.

There is $y_0 \in \mathbf{Y}$ such that

$$\mathrm{Vol}(\omega \cap B_{y_0}(r/2)) \geqslant \frac{\varepsilon}{N} \mathrm{Vol}(B_x(R)) \geqslant \frac{\varepsilon}{N} \mathrm{Vol}(B_{y_0}(r)).$$

In particular, it follows from Proposition 3.3 that there are $C > 0$ and $\alpha \in (0,1)$ such that

$$\|u\|_{\mathrm{L}^2(B_{y_0}(r))} \leqslant C \|u\|^{\alpha}_{\mathrm{L}^2(\omega)} \|w\|^{1-\alpha}_{\mathrm{L}^2(B_x(R') \times [-T,T])}.$$

Let $y \in \mathbf{Y}$ be such that $y \sim y'$. Applying Proposition 3.3, with the smaller set ω' being the ball of radius $r/4$ with same center as the ball of radius $r/2$ in the intersection, we get

$$\begin{aligned} \|u\|_{\mathrm{L}^2(B_y(r))} &\leqslant C \|u\|^{\alpha}_{\mathrm{L}^2(\omega')} \|w\|^{1-\alpha}_{\mathrm{L}^2(B_x(R') \times [-T,T])} \\ &\leqslant C \|u\|^{\alpha^2}_{\mathrm{L}^2(\omega)} \|w\|^{1-\alpha^2}_{\mathrm{L}^2(B_x(R') \times [-T,T])}. \end{aligned}$$

Applying Proposition 3.3 repeatedly from neighbours to neighbours in $\mathbf{G}$ in the same way, we obtain in the end that for any $y \in \mathbf{Y}$,

$$\|u\|_{\mathrm{L}^2(B_y(r))} \leqslant C \|u\|^{\alpha^{\mathrm{diam}(\mathbf{G})}}_{\mathrm{L}^2(\omega)} \|w\|^{1-\alpha^{\mathrm{diam}(\mathbf{G})}}_{\mathrm{L}^2(B_x(R') \times [-T,T])}, \tag{3.5}$$

and since the diameter of $\mathbf{G}$ is bounded by the number of its vertices we obtain our claim after summing (3.5) over all $y \in \mathbf{Y}$. □

Finally, we globalise Proposition 3.4.

Proposition 3.5. *Let $d \geqslant 2$ and $\kappa, R, \varepsilon, T > 0$. There exist $C > 0$ and $\beta \in (0,1)$ so that the following holds.*

For every $(M,g) \in \mathcal{M}(d,\kappa)$, any pairs $u : M \to \mathbb{R}$ and $w : M \times [-T,T] \to \mathbb{R}$ solving

$$\begin{cases} (\Delta_g + \partial_t^2) w = 0 & \text{in } M \times [-T,T] \\ w(y,0) \equiv 0 & \text{and } \partial_t w(y,0) = u(y) \end{cases}$$

and any open set $\omega \subset M$ such that for all $x \in M$,

$$\mathrm{Vol}(\omega \cap B_x(R)) \geqslant \varepsilon \,\mathrm{Vol}(B_x(R))$$

we have the estimate

$$\|u\|_{\mathrm{L}^2(M)} \leq C\|u\|^{\beta}_{\mathrm{L}^2(\omega)}\|w\|^{1-\gamma}_{\mathrm{L}^2(M\times(-T,T))}.$$

Proof. Let $\mathbf{Y}$ be a maximal R-separated set inside M. By construction,

$$M \subset \bigcup_{y\in\mathbf{Y}} B_y(R),$$

which imples, by Proposition 3.4, that there is $C > 0$ and $\beta \in (0,1)$ so that

$$\|u\|_{\mathrm{L}^2(M)} \leq \sum_{y\in\mathbf{Y}} \|u\|_{\mathrm{L}^2(B_y(R))} \leqslant C \sum_{y\in\mathbf{Y}} \|u\|^{\beta}_{\mathrm{L}^2(\omega\cap B_y(R))}\|w\|^{1-\beta}_{\mathrm{L}^2(B_y(R+r_\kappa)\times[-T,T])} \tag{3.6}$$

There is $N > 0$, depending only on d, κ and R, so that for every $y \in \mathbf{Y}$

$$\#\{z \in \mathbf{Y} : B_z(R) \cap B_y(R) \neq \varnothing\} \leqslant N.$$

Indeed, the intersection $B_z(R) \cap B_y(R)$ being non-empty is equivalent to $z \in B_y(2R)$ and, by Proposition 2.5, there is N, independent of y, bounding the cardinality of an R-separated family in $B(y, 2R)$. In particular,

$$\sum_{y\in\mathbf{Y}} \|u\|_{\mathrm{L}^2(\omega\cap B_y(R))} \leq N\|u\|_{\mathrm{L}^2(\omega)} \tag{3.7}$$

and

$$\sum_{y\in\mathbf{Y}} \|w\|_{\mathrm{L}^2(B_y(R+\frac{r_\kappa}{4})\times(-T,T))} \leq N\|w\|_{\mathrm{L}^2(M\times(-T,T))}. \tag{3.8}$$

Inserting (3.7) and (3.8) in (3.6) and applying Hölder's inequality, we obtain finally

$$\begin{aligned}
\|u\|_{\mathrm{L}^2(M)} &\leq C \sum_{y\in\mathbf{Y}} \|u\|^{\beta}_{\mathrm{L}^2(\omega\cap B_y(R))}\|w\|^{1-\beta}_{\mathrm{L}^2(B_y(R)\times(-T,T))} \\
&\leq C \left(\sum_{y\in\mathbf{Y}} \|u\|_{\mathrm{L}^2(\omega\cap B_y(R))}\right)^{\beta} \left(\sum_{y\in\mathbf{Y}} \|w\|_{\mathrm{L}^2(B_y(R+r_\kappa)\times(-T,T))}\right)^{1-\beta} \\
&\leq CN\|u\|^{\beta}_{\mathrm{L}^2(\omega)}\|w\|^{1-\beta}_{\mathrm{L}^2(M\times(-T,T))}.
\end{aligned}$$

□

4. Proof of the main theorems

Proof of Theorem 1.3. Let $M \in \mathcal{M}(d,\kappa)$, $R, \varepsilon > 0$ and let $\omega \subset M$ be an (ε, R)-thick set, and $u \in \mathrm{L}^2(M)$. Recall that μ is the spectral measure for $\sqrt{-\Delta_g}$ and for $\Lambda > 0$, Π_Λ is the spectral projection on $[0,\Lambda]$, that is

$$\Pi_\Lambda u = \int_0^\Lambda \mathrm{d}\mu(\lambda)u.$$

Define $w_\Lambda : M \times [-T,T]$ via the expression

$$w_\Lambda(x,t) = \int_0^\Lambda \frac{\sinh(\lambda t)}{\lambda}[\,\mathrm{d}\mu(\lambda)u](x).$$

By construction,

$$(\Delta_g + \partial_t^2) w_\Lambda = 0 \qquad w_\Lambda|_{t=0} = 0 \qquad \partial_t w_\Lambda|_{t=0} = \Pi_\Lambda u,$$

so that applying Proposition 3.5, we obtain

$$\|\Pi_\Lambda u\|_{\mathrm{L}^2(M)} \leq \|\Pi_\Lambda u\|^{\gamma}_{\mathrm{L}^2(\omega)} \|w_\Lambda\|^{1-\gamma}_{\mathrm{L}^2(M\times(-T,T))}. \tag{4.1}$$

For any bounded measurable function $\varphi \in \mathrm{L}^\infty(\mathbb{R}^+)$, the functional calculus yields the estimate

$$\|\varphi\left(\sqrt{-\Delta_g}\right)\Pi_\Lambda u\|_{\mathrm{L}^2(M)} \leq \|\varphi\|_{\mathrm{L}^\infty([0,\Lambda])} \|\Pi_\Lambda u\|_{\mathrm{L}^2(M)}.$$

Since $\|\sinh(\lambda\cdot)\|_{\mathrm{L}^\infty(-T,T)} \leqslant e^{\lambda T}$, we have that

$$\|w_\Lambda\|_{\mathrm{L}^2(M\times[-T,T])} \leqslant e^{\Lambda T} \|\Pi_\Lambda u\|_{\mathrm{L}^2(M)}.$$

Inserting this into (4.1) gives us

$$\|\Pi_\Lambda u\|_{\mathrm{L}^2(M)} \leqslant C e^{\frac{T}{\gamma}\Lambda} \|u\|_{\mathrm{L}^2(\omega)},$$

in other words (Spec) holds for ω. □

Proof of Theorem 1.7. Letting $v \in \mathrm{L}^2(M)$, one has

$$e^{s\Delta_g} v = \int_0^\infty e^{-s\lambda^2} \mathrm{d}\mu(\lambda) v.$$

Now, for $s > 0$, if we let

$$w_s(x,t) = \int_0^\infty e^{-s\lambda^2} \frac{\sinh(\lambda t)}{\lambda} [\mathrm{d}\mu(\lambda) v](x)$$

one has again $w_s \in \mathrm{W}^{1,2}(M \times [-T,T])$ with

$$\|w_s\|_{\mathrm{L}^2(M\times[-T,T])} \leq C(s,T) \|v\|_{\mathrm{L}^2(M)};$$

essentially, the quadratic frequency decay from the heat equation beats the growth of sinh.

In particular, applying Proposition 3.5, we obtain

$$\|e^{s\Delta_g} v\|_{\mathrm{L}^2(M)} \leq C(s) \|e^{s\Delta_g} v\|^{\gamma}_{\mathrm{L}^2(\omega)} \|v\|^{1-\gamma}_{\mathrm{L}^2(M)},$$

and the proof is complete. □

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